



Assiut
University

Faculty of Computers and
Information
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Course: Computer Graphics
Code: CS351



Question Bank أسئلة إسترشادية

True or False?

1. The volume of a small cube change when its vertices are rotated.
2. The projection preserves straight lines.
3. The dot product of any two perpendicular vectors equals 0
4. With color tracking, you can tell OpenGL to set material properties by only calling glColor.
5. $\text{scale}(z, 3) * \text{scale}(z, 4) = \text{scale}(z, 4) * \text{scale}(z, 3)$

Choose the correct answer for the following questions:

6. What is the correct order of operations of the OpenGL graphics pipeline?
 - a) modelview transformation, projection transformation, divide by w, viewport transform
 - b) projection transformation, modelview transformation, divide by w, viewport transform
 - c) modelview transformation, divide by w, projection transformation, viewport transform
 - d) modelview transformation, viewport transform, divide by w, projection transformation
7. Given $U = (1, 2, -1)$ and $V = (0, -1, 3)$, the cross-product $U \times V$ equals
 - a) (0,-2,-3)
 - b) (5,-3,-1)
 - c) (1, 1,2)
 - d) (5,3,-2)
8. The process of backface..... makes rendering objects quicker and more efficient by reducing the number of polygons for the program to draw.
 - a) FOV
 - b) Culling
 - c) Antialiasing
 - d) Mipmapping

9. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glScalef (-1.0,0.5, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 & -1 & 0 \\ 0 & 0.5 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C
d) D

10. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(-90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C
d) D

11. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(-90.0, 0, 0, 1.0);
glRotatef(90.0, 0, 0, 1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
b) B
c) C

d) D

12. Write down the Modelview matrix after executing the following OpenGL code segment.

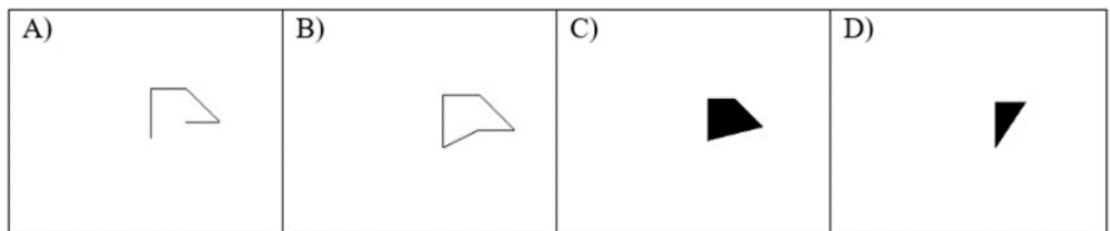
```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-1.0, 2.0, -1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- a) A
- b) B
- c) C
- d) D

13. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
glVertex2f(2.0, 0.0);  
glVertex2f(2.0, 3.0);  
glVertex2f(4.0, 3.0);  
glVertex2f(6.0, 1.0);  
glVertex2f(4.0, 1.0);  
glEnd();
```

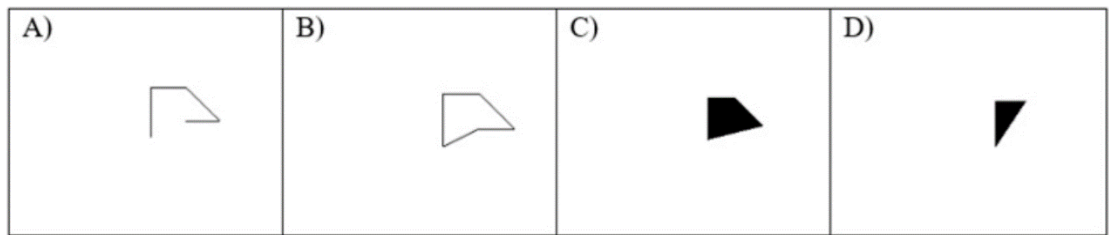


- a) A
- b) B
- c) C
- d) D

14. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
glVertex2f(2.0, 0.0);  
glVertex2f(2.0, 3.0);  
glVertex2f(4.0, 3.0);  
glVertex2f(6.0, 1.0);  
glVertex2f(4.0, 1.0);
```

glEnd();



- a) A
- b) B
- c) C
- d) D

15. Select the output of executing the following OpenGL code segment:

```
glBegin(GL_QUADS);  
  glVertex2f(2.0, 0.0);  
  glVertex2f(2.0, 3.0);  
  glVertex2f(4.0, 3.0);  
  glVertex2f(6.0, 1.0);  
  glVertex2f(4.0, 1.0);  
glEnd();
```

- a) A
- b) B
- c) C
- d) D

16. The OpenGL function used to set up a parallel view volume is

- a) glutCreateWindow
- b) glRectf
- c) glOrtho
- d) glViewport

17. The following two code segments are equivalent:

Code#1

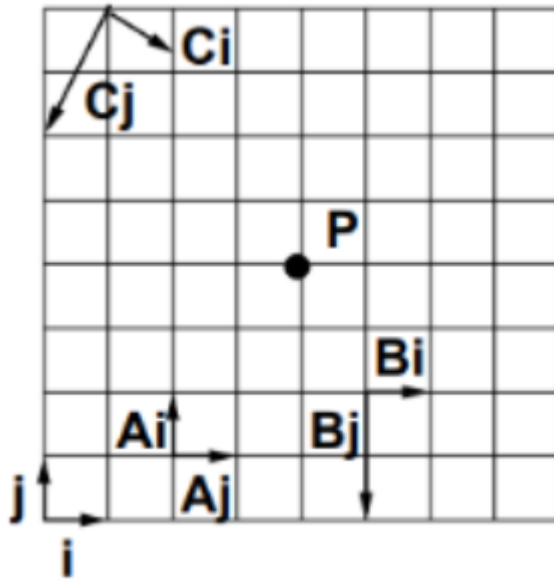
```
glScalef(1.0, 1.0, 2.0);  
glScalef(1.0, 1.0, 4.0);  
dawShape_3();
```

Code#2

```
glScalef(1.0, 1.0, 4.0);  
glScalef(1.0, 1.0, 2.0);  
dawShape_3();
```

- a) True
- b) False

The figure contains a point coordinate P that can be expressed as $P = 4*i + 4*j$, where i and j are basis vectors of unit length along the x and y axes, respectively. Using the figure below to answer the following questions.

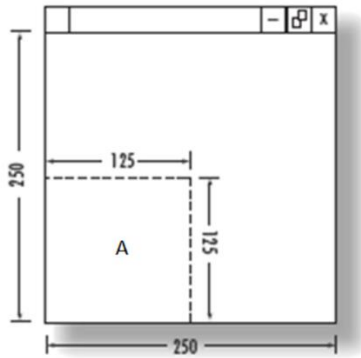


18. is the coordinates of point P with respect to coordinate frames A
 - a) $(3,2)$
 - b) $(2,3)$
 - c) $(4,4)$
 - d) $(-2,3)$

19. is the coordinates of point P with respect to coordinate frames B
 - a) $(1,2)$
 - b) $(-2,-2)$
 - c) $(2,1)$
 - d) $(-1,-1)$

20. The OpenGL command `glClearColor(0.0f, 0.0f, 1.0f, 1.0f)` clear a window with blue color
 - a) True
 - b) False

21. To map the output of the the OpenGL code to the area A (as shown in the figure), we use the .



- a) `glViewport(0, 0, 125, 125);`
- b) `glViewport(125, 125, 125, 125);`
- c) `glViewport(125, 125, 0, 0);`
- d) `glViewport(0, 125, 0, 125);`

22. Given three vertices $P1=(1, 1, 0.5)$, $P2=(1.5, 0.5, 0.5)$, $P3=(1, 0.5, 1)$ on a plane, the equation of the plane is

- a) $x+y+0.5z=2.25$
- b) $x+2y+2z=3.5$
- c) $x+2y+z=3$
- d) $x+y+z=2.5$

23.

```
#define X .525731112119133606
#define Z .850650808352039932

static GLfloat vdata[12][3] = {
    {-X, 0.0, Z}, {X, 0.0, Z}, {-X, 0.0, -Z}, {X, 0.0, -Z},
    {0.0, Z, X}, {0.0, Z, -X}, {0.0, -Z, X}, {0.0, -Z, -X},
    {Z, X, 0.0}, {-Z, X, 0.0}, {Z, -X, 0.0}, {-Z, -X, 0.0}
};

static GLint tindices[20][3] = {
    {0,4,1}, {0,9,4}, {9,5,4}, {4,5,8}, {4,8,1},
    {8,10,1}, {8,3,10}, {5,3,8}, {5,2,3}, {2,7,3},
    {7,10,3}, {7,6,10}, {7,11,6}, {11,0,6}, {0,1,6},
    {6,1,10}, {9,0,11}, {9,11,2}, {9,2,5}, {7,2,11} };

```

The OpenGL code to render the icosahedron is

```
A)
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
        glVertex3fv(&vdata[tindices[i][0]][0]);
        glVertex3fv(&vdata[tindices[i][1]][0]);
        glVertex3fv(&vdata[tindices[i][2]][0]);
    glEnd();
}
```

B)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
    glVertex3fv(&tindices[vdata[i][0]][0]);
    glVertex3fv(&tindices[vdata[i][1]][0]);
    glVertex3fv(&tindices[vdata[i][2]][0]);
    glEnd();
}
```

C)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
    glVertex3fv(tindices[&vdata[i][0]][0]);
    glVertex3fv(tindices[&vdata[i][1]][0]);
    glVertex3fv(tindices[&vdata[i][2]][0]);
    glEnd();
}
```

D)

```
for (i = 0; i < 20; i++) {
    glBegin(GL_TRIANGLES);
    glVertex3fv(vdata[&tindices[i][0]][0]);
    glVertex3fv(vdata[&tindices[i][1]][0]);
    glVertex3fv(vdata[&tindices[i][2]][0]);
    glEnd();
}
```

- a) A
- b) B
- c) C
- d) D

24. The OpenGL command(s) that makes the transformation of Figure A to Figure B:

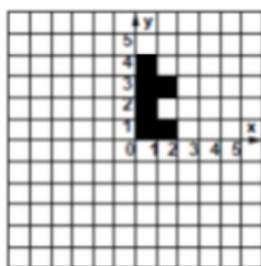


Figure: A

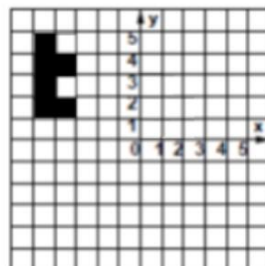


Figure: B

- a) `glTranslatef(-5.0, 0.0, 1.0);`
- b) `glTranslatef(-5.0, 1.0, 0.0);`
- c) `glRotatef(90.0, 0.0, 0.0, 1.0); glTranslatef(-1.0, 1.0, 0.0);`
- d) `glTranslatef(-1.0, 1.0, 0.0); glRotatef(90.0, 0.0, 0.0, 1.0);`

25. The normal vector for the plane $2x+2y+z=1$ is:
- a) $(1,1,1/3)$
 - b) $(0,1,0)$
 - c) $(1/3,2/3,1/3)$
 - d) $(2/3,2/3,1/3)$
26. $\text{Translate}(1,2,3) * \text{Rotate}(90.0, 0, 0, 1.0) = \text{Rotate}(90.0, 0, 0, 1.0) * \dots\dots\dots$
- a) $\text{Translate}(2,-1,3)$
 - b) $\text{Translate}(1,2,3)$
 - c) $\text{Translate}(1,3,-2)$
 - d) $\text{Translate}(-2,1,3)$
27. $\dots\dots\dots$ light component comes from one direction and once it hits a surface, it is scattered.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
28. $\dots\dots\dots$ light comes from one direction and tends to bounce off the surface in a preferred direction.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
29. $\dots\dots\dots$ light is so scattered; it is impossible to tell what its original source was.
- a) Diffuse
 - b) Specular
 - c) Ambient
 - d) Spot
30. $\dots\dots\dots$ projection preserves parallel lines.
- a) Orthographic
 - b) Perspective
 - c) Frustum
 - d) Symmetric

31. The dot product of any two perpendicular vectors equals

- a) 0
- b) 1
- c) (0,0,0)
- d) (1,1,1)

32. If the GL_SPOT_CUTOFF is, spotlight effect equals one.

- a) 0
- b) 45
- c) 90
- d) 180

33. Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glPushMatrix();
glScalef (-1.0,0.5, -1.0);
glRotatef(-90.0, 0, 0, 1.0);
glPopMatrix();
glRotatef(90.0, 0, 0, 1.0);
glScalef(-1.0, 2.0,-1.0);
```

A)	B)	C)	D)
$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

34. The RGB of A cyan color is

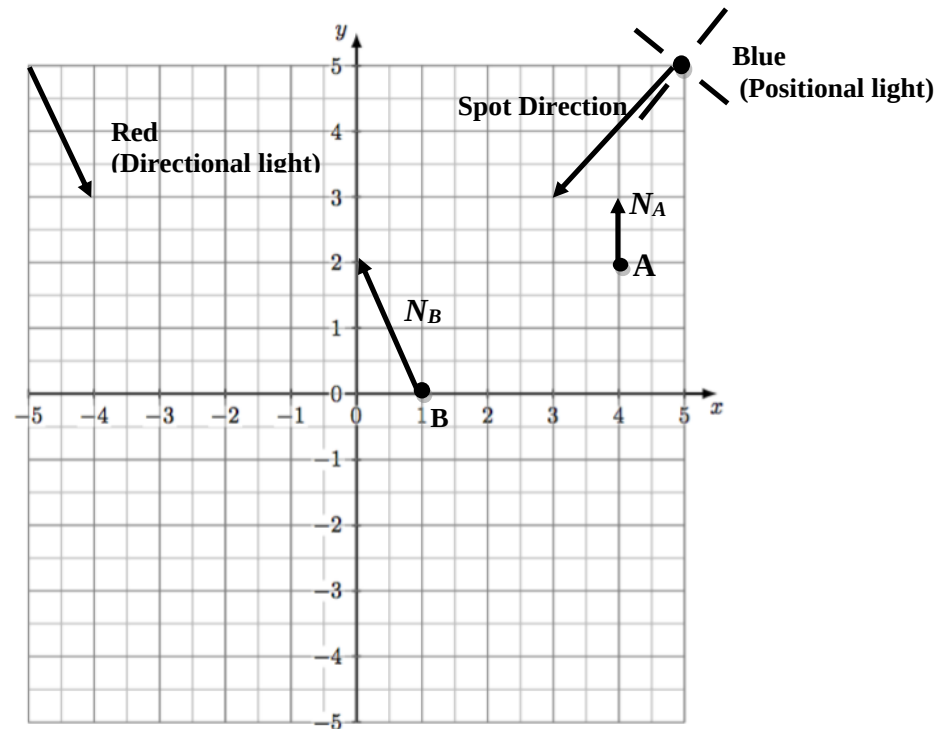
- a) (1.0,1.0,0.0)
- b) (1.0,0.0,1.0)
- c) (0.0,1.0,1.0)
- d) (1.0,1.0,1.0)

35. Convert the following Target Camera to Free Camera

Center = (1, -1, 0)
 Eye = (2, 2, 2)
 Up = (0, 1, 0)

A)	B)	C)	D)
Center = (3, 1, 2) Eye = (2, 2, 2) Up = (0, 1, 0)	Foward = (-1, -3, -2) Eye = (2, 2, 2) Up = (0, 1, 0)	Center = (1, -1, 0) Foward = (-1, -3, -2) Up = (0, 1, 0)	Center = (1, -1, 0) Foward = (-2, -3, -1) Up = (0, 1, 0)

The figure below contains two light sources and two points in 2D. The left light source emits red diffuse light = (1.0, 0.1, 0.1) and its spotlight effect=1. The right light emits blue diffuse light = (0.1, 0.2, 1.0), spot cutoff = 20, and spot exponent=2.0. The normal vectors are shown in the figure. Assume the constant attenuation=0, linear attenuation =2.0, quadratic attenuation=0, the material ambient = (0, 0, 0), the material emission=(0.1,0.1,0.1), the diffuse components of the material = (0.6,0.8,0.4), and the specular components of the material = (0, 0, 0) for the two points. Using the Phong reflection model, answer the following questions:



36. The normalized normal vector N_A =

A) (0,1)	B) (1,0)	C) (0,-2)	D) (-2,0)
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37. The normalized normal vector N_B =

A) $\left(\frac{-1}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$	B) $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$	C) $\left(\frac{-2}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$	D) $\left(\frac{-2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$
---	---	---	---

38. The diffuse intensity for the point A for the blue light source =

A) $\frac{1}{2\sqrt{10}}$	B) $\frac{3}{\sqrt{5}}$	C) $\frac{2}{\sqrt{5}}$	D) $\frac{3}{\sqrt{10}}$
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39. The attenuation factor for the point A for the blue light source =

A) $\frac{1}{\sqrt{10}}$	B) 1	C) $\frac{1}{2\sqrt{41}}$	D) $\frac{1}{2\sqrt{10}}$
--------------------------	------	---------------------------	---------------------------

40. The attenuation factor for the point B for the red light source =

A) $\frac{1}{\sqrt{10}}$	B) $\frac{1}{2\sqrt{41}}$	C) 1	D) $\frac{1}{2\sqrt{10}}$
--------------------------	---------------------------	------	---------------------------

41. The spotlight effect for the blue light for point A =

A) 0	B) 1	C) $\frac{81}{82}$	D) $\frac{82}{81}$
------	------	--------------------	--------------------

42. The spotlight effect for the blue light for point B =

A) 0	B) 1	C) $\frac{81}{82}$	D) $\frac{82}{81}$
------	------	--------------------	--------------------